Solar Dark J–V Analyzer

Quick Guide to Region Selection and Nonlinear Fitting

SutharRakesh • Single- and two-diode models • GaAs and perovskite measurements

Purpose.
A practical reference for choosing input settings, initial-estimate regions, fitting models, and acceptable numerical bounds.
Fit each dark J–V dataset at its *measured* temperature (e.g., 83–300 K).

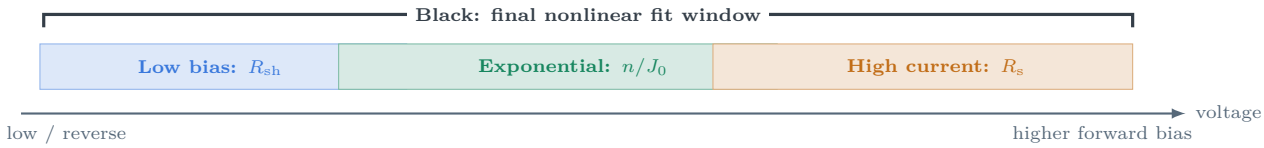
1 Prepare a measurement

1. **Load two columns:**
   Voltage V (V), then current density J or current I.
   Upload `.txt`, `.csv`, `.tsv`, or `.dat`; paste values; or select the synthetic example.
2. **Set units correctly:**
   `V + J (mA/cm^2)`, `V + J (A/cm^2)`, or `V + I (A)`.
3. **Check signs:**
   The app assumes positive forward voltage produces positive dark current.
   Use *Flip voltage sign* / *Flip current sign* only if needed.
4. **Set temperature (K)**
   for this particular measurement; choose a diode model and a fitting objective.
   Treat separate hysteretic sweep directions separately.

Upload limits: file ≤ 5 MB, 8–5000 valid data points, and a nonzero voltage span. The final fit window also requires at least 8 points.

2 Four colored fitting regions

Region in app	Select this part of the data	What it controls
Blue: R_{sh}	Approximately ohmic, low-bias portion; a small reverse-bias interval can help. Avoid breakdown/nonlinear leakage.	Initial estimate from $\left(dJ/dV\right)^{-1}$.
Green: n/J_0	Positive-current, forward-bias exponential interval: nearly straight on $\ln J_{diode}$ vs corrected voltage.	Initial ideality-factor and saturation-current estimates.
Orange: R_s	High forward-current interval where the series-voltage drop is detectable. Exclude instrument compliance/plateaus.	Initial series-resistance estimate.
Black: Final fit	All valid points in the range where the selected circuit model should apply.	**Only this window** sets the points in final nonlinear optimization.



Conceptual windows only: locations and overlaps depend on the measurement; these are *not* fixed voltage limits.

Key distinction:
Blue, green, and orange give *starting guesses*; their region-wise estimates are not the final reported fit.
Click **AUTO SUGGEST REGIONS**, inspect all four shaded windows, adjust when needed, then click **RUN NONLINEAR FIT**.

3 After the fit: verify and export

- Compare **J–V**, **Log $|J|$–V**, and **Residual** tabs.
  The log plot uses magnitudes for negative current; it is *not* the same as the positive-only log-fitting objective.
- Watch for systematic residual trends, parameter bounds, poor convergence, current-compliance plateaus, and overlapping scan directions.
- Report the selected voltage window, temperature, fit objective, model, and any fixed parameters.
  High R^2 alone does not establish unique parameters.
- Download **DATA FOR ORIGIN** and **PARAMETERS** (TSV) for figures and record keeping.

Model Choice, Limits, and Fit Checks

Use this page while comparing candidate fits for a single measurement.

4 Models available in the application

Define the internal diode voltage

$$U = V - JR_{\mathrm{s}},$$

thermal voltage $V_{\mathrm{T}} = k_{\mathrm{B}}T/q$, and current density J in $\mathrm{A/cm^2}$.

Single diode: $\mathrm{J} = \mathrm{J}_0 \left[\exp\left(\frac{U}{nV_{\mathrm{T}}}\right) - 1\right] + \frac{U}{R_{\mathrm{sh}}}$

Two diode: $\mathrm{J} = \mathrm{J}_{01} \left[\exp\left(\frac{U}{n_1 V_{\mathrm{T}}}\right) - 1\right]$
$+ J_{02} \left[\exp\left(\frac{U}{n_2 V_{\mathrm{T}}}\right) - 1\right] + \frac{U}{R_{\mathrm{sh}}}$

R_{s} and R_{sh} are *area-normalized* resistances ($\Omega\,\mathrm{cm^2}$); J_0, J_{01}, J_{02} are saturation-current densities ($\mathrm{A/cm^2}$).

Selection	Practical decision rule
Single diode	Start here; use if the curve and residuals are adequately described by one exponential contribution plus resistances.
Two diode	Try when single-diode residuals show systematic structure and the data support two distinct contributions; inspect parameter stability.
Fixed $n_1 = 1, n_2 = 2$	Available option within two-diode fitting; use only when this assumption is justified for the device.
Other mechanisms	Tunneling, contact-limited transport, and breakdown are *not separate fitted models* in this app.

5 Actual numerical parameter bounds in the code

Parameter(s)	Lower	Upper	Unit
J_0, J_{01}, J_{02}	10^{-20}	10^{-1}	$\mathrm{A/cm^2}$
n, n_1, n_2	0.5	6.0	dimensionless
R_{s}	10^{-10}	10^{6}	$\Omega\,\mathrm{cm^2}$
R_{sh}	10^{-1}	10^{14}	$\Omega\,\mathrm{cm^2}$

Numerical bounds ≠ physical validity.
An n near 1 or 2 can be consistent with familiar diode limits *under suitable assumptions*; neither value proves a unique mechanism.
Values outside 1–2 (but within 0.5–6) require closer scrutiny, especially for complex devices.
Parameters close to any bound may be poorly constrained.

6 Fitting controls and interpretation

Control	When to choose / what to check
Balanced (asinh)	Good default across many current decades; handles positive and negative J.
Linear J–V	Targets absolute-current errors; high-current data may dominate.
Log J–V	Uses *only positive measured currents*; reverse and zero-current points do not enter this objective.
Noise-aware	Use an experimentally justified absolute noise floor and relative uncertainty.
Multi-start attempts	Tries different starting points (1–30 attempts); more starts may improve robustness but take longer.
Fix selected parameters	Lock a parameter only when independently known; fixing the wrong value biases the fit.
Check parameter reliability	Inspect warnings about parameter correlation, sensitivity, or alternate nearly equivalent fits.

Source code and updates:
github.com/SutharRakesh/solar-dark-jv-analyzer .